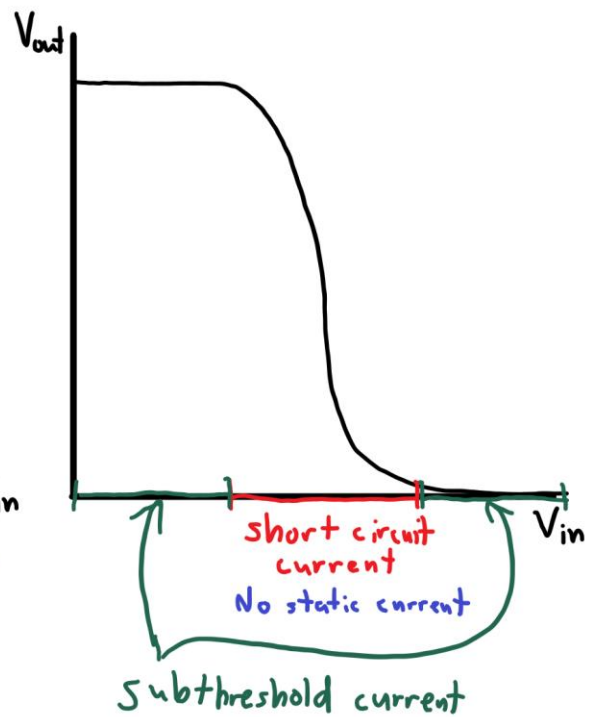
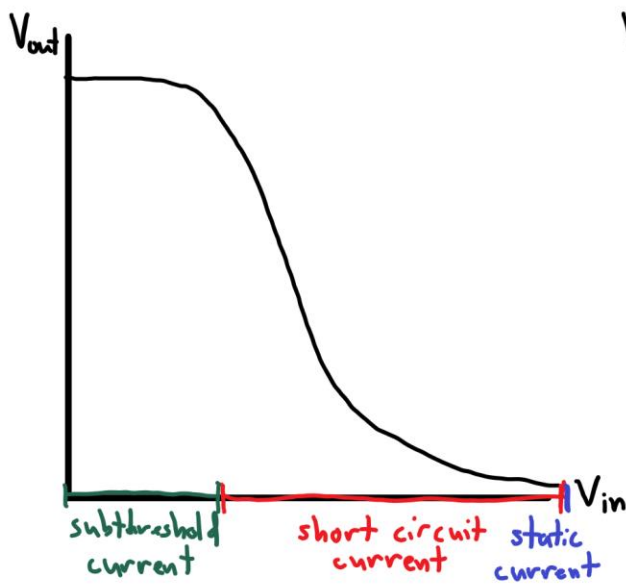
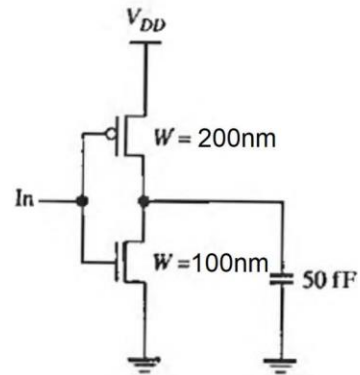
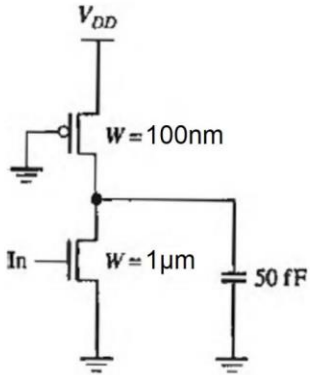


1a)



1b)

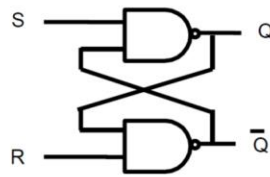
P_D same for both since both have same C_L, α, f, V_{DD}

$$P_D = \alpha f C V_{DD}^2 = 1 \cdot 2 \cdot 10^9 \cdot 50 \cdot 10^{-15} \cdot 1.2^2$$

$$P_D = 144 \mu A$$

2]

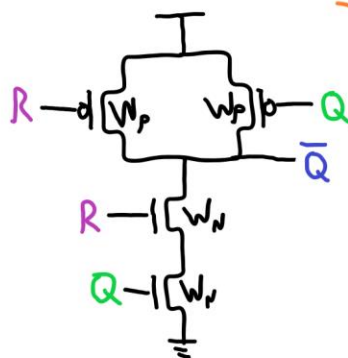
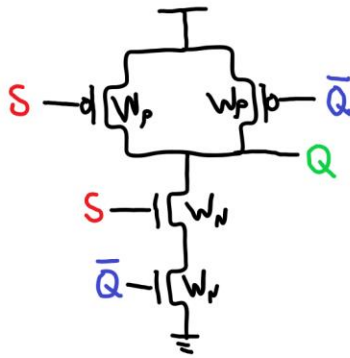
NAND SR-Latch



$$\tau_{ps\bar{a}} = 400\text{ps}, \tau_{psa} = 200\text{ps}, \tau_{qa\bar{a}} = 200\text{ps}$$

$$\tau_{pra} = 400\text{ps}, \tau_{pr\bar{a}} = 200\text{ps}, \tau_{p\bar{a}a} = 200\text{ps}$$

$$L = 100\text{nm}$$



All W shorthand for $\frac{W}{L}$

For pull-up,

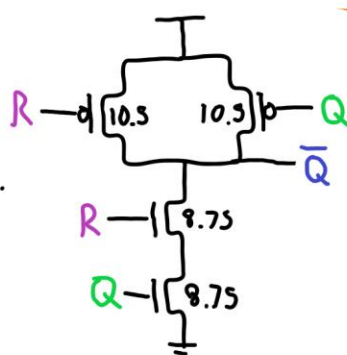
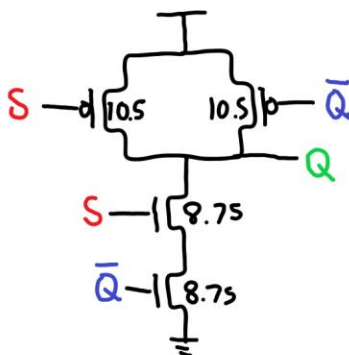
$$\tau_p = 0.7 R_{effp} C = 0.7 \cdot 30k \cdot \frac{L}{W_p} \cdot 100f = 200p$$

$$\frac{W_p}{L} = 10.5$$

For pull-down,

$$\tau_{pHL} = 0.7 R_{effn} C = 0.7 \cdot 2 \cdot 12.5k \cdot \frac{L}{W_n} \cdot 100f = 200p$$

$$\frac{W_n}{L} = 8.75$$



All W shorthand for $\frac{W}{L}$